Inch - Pound

MIL-M-80033D
21 April 1992
SUPERSEDING
MIL-M-80033C
8 March 1985

MILITARY SPECIFICATION

MILLING MACHINES, COMBINATION HORIZONTAL AND VERTICAL, KNEE TYPE

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers knee type, combination horizontal and vertical milling machines, electric motor driven, floor mounted, with plain or universal table mounting, with equipment and accessories specified in Section 3. The requirements set forth in this specification and the completed ordering data (paragraph 6.2) establish the Government's minimum requirements for each procurement.

1.2 Classification. This specification covers milling machines of the following types, styles, and sizes. The type, style and size to be furnished shall be as specified (see 3.5 and 6.2).

Type I - Dual spindle design

Style A - Turret positioning of horizontal and vertical spindle mounted on opposite ends of the ram

Size 1 - 21 inches minimum longitudinal table travel
Size 2 - 22 inches minimum longitudinal table travel
Size 3 - 28 inches minimum longitudinal table travel
Size 4 - 38 inches minimum longitudinal table travel

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Defense Industrial Plant Equipment Center, ATTN: DIPEC-SSM, 2163 Airways Boulevard, Memphis, Tennessee 38114-5051, by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 3417

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Style B - Ram mount vertical spindle, integral horizontal spindle

Size 1 - 30 inches minimum longitudinal table travel
Size 2 - 34 inches minimum longitudinal table travel

Type II - Single spindle design

Style A - Single swivel parallel to ram movement, dual ram for arbor support while horizontal milling

Size 4 - 34 inches minimum longitudinal table travel

Style B - Dual swivel spindle mounted on front of ram, arbor support or ram near column for horizontal milling

Size 1 - 22 inches minimum longitudinal table travel

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

MILITARY

MIL-M-18058 - Machinery, Metal and Woodworking, Packaging of.

STANDARDS

FEDERAL

FED-STD-H28 - Screw Thread Standards for Federal Services.

FED-STD-376 - Preferred Metric Units for General Use by the Federal Government.

MILITARY

MIL-STD-129 - Marking for Shipment and Storage

MIL-STD-461 - Electromagnetic Emission and Susceptibility Requirements for the Control of Electromagnetic Interference.

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MIL-STD-462 - Electromagnetic Interference Characteristics, Measurement of.

MIL-STD-1190- Minimum Guidelines For Level C Preservation, Packing and Marking.

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Document Order Desk, Building 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

CODE OF FEDERAL REGULATIONS (CFR)

U. S. DEPARTMENT OF LABOR, OCCUPATIONAL SAFETY AND HEALTH ADMINISTRATION (OSHA)

29 CFR 1910 - Occupational Safety and Health Standards.

(Application for copies should be addressed to the Superintendent of Documents, U. S. Government Printing Office, Washington, DC 20402-0001.)

(Copies of specifications, standards, handbooks, drawings, publications, and other Government documents required by contractors in connection with a specific acquisition should be obtained from the contracting activity.

2.2 Non-Government publications. The following document(s) form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

AMERICAN GEAR MANUFACTURERS' ASSOCIATION (AGMA)

AGMA 390.03a - Gear Classification, Materials and Measuring Methods for Bevel, Hypoid, Fine Pitch Wormgearing and Racks Only as Unassembled Gears. (DOD adopted)

AGMA 2000-A88 - Gear Classification and Inspection Handbook - Tolerances and Measuring Methods for Unassembled Spur and Helical Gears (including Metric Requirements). (DOD adopted)

AGMA 2001-B88 - Fundamental Rating Factors and Calculation Methods for Involute Spur and Helical Gear Teeth. (DOD adopted)

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(Application for copies should be addressed to the American Gear Manufacturers' Association, Standards Department, 1500 King Street, Suite 201, Alexandria, VA 22314-2717.)

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

ANSI B5.1M - T-Slots, Their Bolts, Nuts, Tongues, and Cutters. (DOD adopted)

ANSI B5.10 - Machine tapers. (DOD adopted)

ANSI B5.18 - Spindle Noses and Tool Shanks for Milling Machines. (DOD adopted)

ANSI B11.8 - Drilling, Milling and Boring Machines, Safety Requirements for the Construction, Care and Use of. (DOD adopted)

ANSI B93.114M - Pneumatic Fluid Power Systems Standard for Industrial Machinery. (DOD adopted)

(Application for copies should be addressed to the American National Standards Institute, ATIN: Sales Department, 1430 Broadway, New York, NY 10018-3363.)

NATIONAL ELECTRICAL MANUFACTURERS' ASSOCIATION (NEMA)

NEMA ICS 1 - Industrial Controls and Systems. (DOD adopted)

NEMA ICS 3 - Industrial Systems. (DOD adopted)

NEMA MG 1 - Motors and Generators. (DOD adopted)

(Application for copies Enclosures be sent to the National Electrical Manufacturers' Association, 2101 L Street, N.W. Washington, DC 20037-1580).

NATIONAL FLUID POWER ASSOCIATION (NFPA)

NFPA T2.24.1 - Hydraulic Fluid - Power System Standard for Stationary Industrial Machinery. (DOD adopted)

(Application for copies should be addressed to the National Fluid Power Association, 3333 North Mayfair Road, Milwaukee, WI 53222.)

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

NFPA 79 - Electrical Standard for Industrial Machinery. (DOD adopted)

(Application for copies should be addressed to the National Fire Protection Association, Batterymarch Park, Quincy, MA 02269-9101.)

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INTERNATIONAL ORGANIZATION FOR STANDARDIZATION

ISO - 54 - Cylindrical Gears for General Engineering and for Heavy Engineering - Modules and Diametrical Pitches

(Application for copies should be addressed to the American National Standard Institute, Inc., 1430 Broadway, New York, NY 10018).

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM-D-3951 - Commercial Packaging Standard Practice for (DoD Adopted)

(Application for copies should be addressed to the American Society for Testing and Materials, Sales and Service Department, 1915 Race Street, Philadelphia, PA 19103).

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), one complete milling machine shall be subjected to first article inspection (see 6.4) in accordance with 4.4.

3.2 Design. The milling machine shall be new (not a prototype) and one of the manufacturer's current production models capable of operations in accordance with the requirements herein. The milling machine shall include all components, parts, and features necessary to meet the performance requirements specified herein.

3.2.1 Measuring and indicating device calibrations. Measuring and indicating devices shall be graduated in inch-pound (see 6.5.1), metric (see 6.5.2), or dual (inch-pound and metric) units as specified by the procuring activity (see 6.2). Metric units shall be in accordance with the FED-STD-376. Regardless of the measurement system used, all measuring and indicating devices on the milling machine shall be graduated in the same system.

3.2.1.1 Dials. All dials used to indicate machine axes position or tool movement shall be graduated in increments not larger than 0.0010 inch. Dial and/or band circumferences, on hand wheels, shall be permanently and legibly engraved or etched on a nonglare background with graduations that can be read from the operator's normal position. All feed dials shall have independent

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zero adjustments and shall be calibrated in such a manner that the last dial graduation progresses into and is continuous with the first dial graduation as the dial is rotated through the zero position.

3.2.1.2 Digital readout system. When specified (see 6.2), a digital readout system shall be installed on the X, Y and Z axes of travel. The system shall consist of a position feedback device, an electronic display for each axis, and all related components, mounting hardware, and electrical equipment necessary to ensure complete operation of the system. The system shall provide a readily visible digital display of the full working range of each axis on which installed. The system resolution shall be 0.0001 inch. The repeatability shall be 0.0001 inch, measured over the full travel on each axis. Special requirements for the system components shall be as specified (see 6.2). The display shall indicate axis location in both the inch-pound and metric units. An input power source fluctuation of ± 10 percent shall not adversely affect the system operation.

3.2.1.2.1 Position feedback device. When a digital readout system is specified (see 6.2), a position feedback device including all related components, hardware, and electrical equipment shall be provided for each axis having a readout system. The feedback device shall be of the independent type with a resolution equal to the resolution of the digital readout system. Each feedback device shall be installed with a cover or guard to prevent physical damage to any portion of the feedback device. The connecting cable from the position feedback device shall be shielded to protect against electronic interference and shall be armored to protect against contamination and damage. The device shall provide axis travel feedback over the entire range of axis travel for each specified axis. Installation of the position feedback device shall not limit machine travel on any axis nor eliminate manual machine stops. Provisions shall be made to prevent damage to the feedback device in the event of over travel. The feedback unit and related components shall be protected from contamination. The digital readout system shall be interchangeable between like machines, procured under the same contract, with no structural rework required. Only screw or bolt removal and replacement shall be required, along with reconnecting electrical connections which are plug in type, using existing plugs and connectors, or wire changes to existing terminal blocks.

3.2.1.2.2 Electronic display console. When specified (see 6.2), the electronic display console for the digital readout system shall contain a universal digital display with individual illuminated digits of not less than 0.43 inch in height. Digits shall be high efficiency displays for high light ambient conditions with wide viewing angle and shall be readable from the operator's normal work station. The console shall display at least six digits per axis simultaneously. The console shall also provide keypad entry, memory, absolute/incremental counting, data preset, power failure indicator, inch/metric conversion, and absolute zero recall. The display console shall be installed so as to be readily accessible and easily read from the operator's normal work station. The console shall be mounted to withstand shock and vibration generated by the machine. Each connecting cable shall be armored and shall be securely fastened to the structure of the machine. Location of installed cables shall prevent any interference with normal

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machine operation. The consoles shall be interchangeable between like machines, procured under the same contract, with no structural rework required. Only screw or bolt removal and replacement shall be required, along with reconnecting electrical connections which are plug-in type, using existing plugs and connectors, or wire changes to existing terminal blocks.

3.2.2 Reclaimed materials. The machine may contain reclaimed materials to the maximum extent possible provided such materials will not jeopardize the intended use, performance, or design life of the machine. Reclaimed materials shall have been collected or recovered from solid waste and reprocessed to become a source of raw materials, as opposed to virgin raw materials. Design life of the machine shall be no less than five years of service or not less than 10,000 hours of operation. None of the above shall be interpreted to mean that the use of used, rebuilt, or remanufactured end products are allowed under this specification.

3.2.3 Energy efficiency. The milling machine and its components that directly consume energy in normal operation shall be designed and constructed for energy efficiency in accordance with the developments available within the industry.

3.2.4 Controls. All electrical, mechanical, hydraulic, and pneumatic operating controls shall be located convenient to the operator's work station(s).

3.2.5 Safety and health requirements. All parts, components, mechanisms, and assemblies furnished on the milling machine, whether or not specifically required herein, shall conform to all requirements of OSHA 29 CFR 1910 and ANSI B11.8. If a conflict arises between 29 CFR 1910 and ANSI B11.8, 29 CFR 1910 shall apply. Additional safety and health requirements shall be as specified (see 6.2). Covers, guards, or other safety devices normally furnished as standard on the manufacturer's commercial milling machines supplied to the commercial market shall be provided for the point of operation and all other parts of the milling machine that present safety hazards. If point-of-operation guarding is influenced by workpiece size and configuration, the procuring activity shall provide part design data to the supplier of the machine (see 6.2). All special guarding shall meet OSHA 29 CFR 1910. The safety devices shall not interfere with the operation of the machine. All guards shall allow access to the guarded part(s).

3.2.6 Environmental protection. During the manufacture, service, transportation, or storage of the machine and its components, the machine shall meet all the applicable Environmental Protection Agency (EPA) and OSHA restrictions for materials classified as hazardous to the environment in effect on the date of the contract. The operation of the machine at its final location shall comply with all applicable EPA and OSHA regulations in effect on the date of the contract. These regulations apply to the emission of materials hazardous to the environment or the operator's health. Specific environmental protection exceeding the established EPA and OSHA regulations shall be as specified by the procuring activity (see 6.2).

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3.2.7 Lubrication. Means shall be provided to ensure that failure does not result from lack of lubrication for all moving parts. Mechanisms requiring periodic manual lubrication shall be provided with grease fittings and/or oil cups, which shall be usable without dismantling or removing components from the machine. Means shall be provided for each recirculating lubrication system to ensure that failure does not result from lubricant contaminants. Each reservoir shall have at least 24-hour capacity and a means for determining fluid level. Means shall be provided for determining lubricant level in each reservoir and shall clearly give an indication of a low lubricant level condition to the vantage point of the operator's normal work station. An automatic lubrication system shall be provided for all ways and axis screws. Loss of lubrication to any component serviced by the automatic lubrication system shall automatically alert the operator to a lubrication system malfunction.

3.2.8 Interchangeability. All parts of the end item having the same manufacturer's part number shall be manufactured to definite standards and tolerances that will provide for the interchangeability of respective replacement parts between end items of the same model without modification of the part or the machine. All replacement parts shall be available domestically.

3.3 Construction. The milling machine shall be constructed of new parts, without defects, and free of repairs. The structure shall withstand all forces encountered during operation of the milling machine to its maximum rating and capacity without permanent distortion or failure.

3.3.1 Castings and forgings. All castings and forgings shall be free of defects, scale, and mismatching. No processes such as welding, peening, plugging, or filling with solder or paste shall be used for reclaiming any defective part. Such processes may be used only for enhancing surface finish for appearance.

3.3.2 Fastening devices. All fasteners shall be installed in such a manner as to prevent unintentional change of tightness. Fastening devices subject to removal or adjustment shall remain tight until manually loosened. Fasteners subject to removal or adjustment shall not be swagged, peened. staked or otherwise permanently deformed.

3.3.3 Surfaces. All surfaces shall be clean and free of harmful or extraneous materials. All edges shall be either rounded or beveled unless sharpness is required to perform a necessary function. Except as otherwise specified herein, the condition and finish of all surfaces shall be in accordance with the manufacturer's commercial practice.

3.3.4 Welding, brazing, or soldering. Welding, brazing, or soldering shall be employed only where specified in the original design. Any excess material used for such operations shall be thoroughly removed from the part(s) upon completion of the operations. None of these operations shall be employed as a repair measure for defective parts.

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3.3.5 Painting. Unless otherwise specified (see 6.2), the paint color and application shall be in accordance with the manufacturer's commercial practice and OSHA requirements.

3.3.6 Threads. All threaded parts used on the milling machine and its related attachments and accessories shall conform to FED-STD-H28 and the applicable "Detailed Standard" section referenced therein. Metric units shall be in accordance with the applicable section of FED-STD-H28.

3.3.7 Drive train gears. All gears of spindle and axis drive trains shall conform to or exceed all provisions of AGMA 390.03a, AGMA 2001-B88, and AGMA 2000-A88 for the inch-pound system of measurement or ISO - 54 for the metric system of measurement. The gears shall be of proper width and size to transmit full rated torque and horsepower throughout the speed ranges without failure for the expected service life of the machine. Gears in the drive train(s) shall be hardened and ground steel. Working surface hardness shall be adequate to ensure expected service life of the machine and shall be not less than Rockwell C-48.

3.3.8 Electromagnetic interference control. When specified (see 6.2), equipment furnished under this specification shall comply with MIL-STD-461. The equipment and subsystem(s) class and the emission and susceptibility requirements shall be as specified.

3.4 Components. The milling machine shall consist of not less than a column and base, knee, universal table mount (as applicable), saddle, turret, ram, table, spindle and feed mechanism.

3.4.1 Column and base. The column and base shall be of a design and material of sufficient strength and rigidity for supporting and maintaining mutual component alignment, as necessary for meeting the performance, accuracy and alignment requirements herein. The column shall have machined ways for the precision movement, under maximum cutting load conditions for the rated horse power, along the entire vertical axis of travel. Access doors or removable covers shall be provided for servicing the components located therein. Means for leveling the machine shall be provided. Means for anchoring the machine to the floor or foundation after installation shall be provided.

3.4.2 Knee. The knee shall be of a design and material of sufficient strength and rigidity to maintain the performance, accuracy and alignment requirements specified herein. The knee shall be fitted to the column ways with adjustable gibs and shall be equipped with wipers or other means to prevent dust and chips from entering the ways. Means shall be provided for clamping the knee in any position within its rated travel along the column. The knee elevating screw shall have either square or Acme type threads. If required (see 6.2), all feed screws shall be adequately guarded. The knee adjusting crank shall be of the safety lock type or shall be positively locked or disabled when the power feed or rapid traverse is engaged.

3.4.3 Universal table mount. When specified (see 6.2), the machine shall be supplied with a universal saddle and swivel block that will permit the table to swivel at least 35 degrees right and left of zero degrees position.

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The swivel block shall have a bearing area large enough to support, without distortion, the maximum load of the table in any angular position while performing any type of milling operation basic to the requirements of the milling machines covered by this specification. The swivel block shall have not less than three clamping points and shall be graduated to indicate its angular setting.

3.4.4 Saddle. The saddle shall be of a design and material that will provide sufficient strength and rigidity to maintain the performance, accuracy and alignment requirements herein. The saddle shall have machined ways fitted for precision movement for the entire cross feed axis of travel. The saddle shall be equipped with guides, adjustable gibs and a means of preventing dust and chips from entering the ways. The cross feed handle on machines with power feeds shall have a means for disengagement when power feed or rapid traverse is engaged.

3.4.5 Turret. Type I, Style A machines shall have a turret which will permit the ram to swivel in a horizontal plane. The turret shall be machined and fitted to the top of the column. Means shall be provided for clamping the turret at clearly indicated zero points showing suitable alignment of the ram with the worktable. The top of the turret shall have accurately machined dovetails or square lock ways matching the ways of the ram. The plane of rotation of the turret shall be parallel to the surface of the table.

3.4.6 Ram. The ram shall be rigidly constructed and shall be fitted, as applicable, to matching ways on top of the column or the top of the turret. The ways shall be machined and fitted for precision movement to maintain the performance accuracy and alignment required herein. Means shall be provided for clamping the ram in any position in its rated range of travel. Suitable provisions shall be made for manual positioning of the ram. Movement of the ram shall be parallel to the surface of the worktable.

3.4.7 Table. The table shall be of a design and material that will provide sufficient strength and rigidity to maintain the performance, accuracy and alignment requirements herein. The table shall have machined ways fitted for precision movement for the entire longitudinal axis movement. The table shall be equipped with adjustable gibs and means for preventing dust and chips from entering the ways. The table feed handle on machines with power feed shall have means for disengagement when power feed or rapid traverse is used. The table shall have not less than three T-slots conforming to ANSI B5.1M. The T-slot size shall be no less than 5/8 inches. The front face of the table shall be a finished surface and shall have a slot or slots running the entire length of the table for mounting the adjustable table stops. The table shall have coolant drain channels with chip screens and provisions for returning the coolant to the reservoir. Table size and longitudinal and traverse travel capacities shall be as stated in Table I. When specified (see 6.2) an oversized worktable of the size and travel capacity shall be provided.

3.4.8 Spindle. The spindle shall be of a heat treated alloy steel. The spindle shall utilize precision type roller or ball bearings. Means shall be provided for adjusting the spindle bearings unless a cartridge type system is provided. The spindle nose shall be precision ground, conforming to

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ANSI B5.18 with the spindle nose taper size as specified in Table I for the specific machine size. A spindle braking mechanism shall be provided. Spindle speed changes shall be accomplished by means of levers, knobs or push buttons with an open faced dial or plate to indicate the selected speed. When specified (see 6.2), Type II , Style A, Sizes 1 and 2 machines shall have power feed to the quill.

3.4.9 Feed mechanism. Feed changes shall be accomplished by means of levers or knobs with an open faced dial or plate to indicate the feed rate in engagement. Use of pickoff gears for changes in feed rate shall not be acceptable. The feed control levers shall be directional and shall allow independent reversing of each feed. The feed mechanism shall be equipped with safety overload protection devices. The spindle drive motor and the feed motor shall be interlocked so that power feeds are automatically stopped whenever the spindle is stopped.

3.4.9.1 Leadscrews, gear shaft and bearings. The leadscrews for longitudinal, cross and vertical travel shall be mounted in precision type ball or roller bearings. There shall be at least one ball or roller bearing on each end of the leadscrews. The gear shafts bearings shall be of the precision type ball or roller bearings.

3.4.10 Power rapid traverse. Unless otherwise specified (see 6.2), power rapid traverse independent of feed changes shall be provided for the table in both directions of longitudinal and cross feed. Power rapid traverse is not required on a machine for which no rapid traverse rate is specified in Table I.

3.4.11 Clamping. Lever and/or electrically applied clamping shall be provided for fastening each component relative to a machine axis (X, Y or Z). Applying and releasing the clamps shall be accomplished without measurable movement to the affected component.

3.4.12 Stops. If power operation is required, adjustable and power stops shall be provided for all powered axes.

3.4.13 Coolant system. The coolant system shall be of the flood type. The coolant recycling system shall include a sump or reservoir, an electrically powered pump and all necessary piping. The sump or reservoir shall be of sufficient capacity to permit the continuous flow of coolant while maintaining at least seven gallons of coolant in the sump or reservoir. The system shall have a means for draining and cleaning. The system shall include a strainer which can be removed for cleaning without the removal of mill components other than a strainer cover. A valve and flexible metal hose, conveniently located to the machine operator's normal working position, shall be provided to allow the operator to direct and control the amount of coolant flow.

3.4.14 Pneumatic system. If a pneumatic system is supplied, it shall meet or exceed the requirements of ANSI B93.114M.

3.4.15 Hydraulic system. If a hydraulic system is supplied, it shall meet or exceed the requirements of NFPA T2.24.1. The hydraulic system shall be

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complete, including, but not limited to pump(s), valves, piping, cylinders, pressure controls and all other components necessary to make the system completely functional.

3.4.16 Electrical requirements. The machine shall be wired to conform to NFPA 79. Each machine shall be wired to draw all of its electrical power from a single circuit. Unless otherwise specified (see 6.2), the machine shall be designed to operate on dual 230/460-volt, 3-phase, 60-Hertz (Hz) input power. Conversion from one voltage to the other shall be accomplished by the reconnection of existing components. The machine shall be initially wired for operation on the voltage specified by the procuring activity (see 6.2).

3.4.16.1 Motors. Shall be rated for continuous duty and shall have ball or roller bearings of the sealed and permanently lubricated type. All motors shall conform to the requirements of NEMA MG-1. Unless otherwise specified (see 6.2), motors shall be alternating current (AC), 60-Hertz (Hz), or direct current (DC) industrial type motors with dripproof enclosures. Each motor control shall conform to the requirements of NEMA ICS-1 and NEMA ICS-3.

3.4.16.2 Work light. When specified (see 6.2), a work light shall be provided. The light shall have a protective shield, an on-off switch, and shall be adjustable to reflect maximum light on the work area. The light shall be rated at no less than 100 watts. If a built-in lighting system is provided, the adjustable light described above shall not be required.

3.4.16.3 Hour meter. When specified (see 6.2), the milling machine shall be equipped with an hour meter. The hour meter shall be installed to display accumulated operating time of the main drive motor. The meter shall be of the nonresetting type and shall have a range of 0 to 99,999 hours in increments of not greater than 1 hour. The meter shall be sealed to prevent the entrance of dust and moisture and shall be mounted to withstand shock and vibration generated by the machine.

3.4.16.4 Identification of electrical circuits. The conductors in the electrical system shall be identified at each termination to correspond with the identification on the wiring diagrams and schematics. Conductors shall be color coded and identified in accordance with NFPA 79. The electrical system/schematics provided with each machine shall be an exact duplication of the electrical system and wiring in the mill. Diagrams/schematics shall clearly indicate the individual, point-to-point, wire and wire termination point locations in the electrical system.

3.4.17 Machine ways. The machine ways shall be adequately protected to prevent chips, dirt, coolant and other hazards from causing damage or inaccurate functioning of any interfacing surface members of the machine. The machine ways shall be hardened to not less than Rockwell C-42. The finish on the machine ways shall not be more than 32 micro-inches. Mating ways of the knee, saddle and table shall be provided with adjustable gibs. Wear strips shall not be used.

3.5 Size and capacity. Unless otherwise specified (see 6.2), the size and capacity of the machine shall meet the requirements of Table I for the

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machine specified. Machine capacity shall be not less than the stated requirements of the size ordered (see Table I, NOTE:). When a range is given, the required performance shall be from the stated minimum or less to the stated maximum or greater.

3.5.1 Physical size limitations. Physical size of the machine shall be limited, when specified (see 6.2). The physical size of the machine shall not exceed the height, width, and length restrictions specified by the procuring activity to ensure the machine will fit in its future operating location.

3.6 Alignment tolerances. Each milling machine shall meet the alignment tolerances specified in Table II.

3.7 Performance. The milling machine shall be capable of satisfactory performance when tested in accordance with Section Four.

3.7.1 Additional performance requirements. Performance features required in addition to, or in lieu of, the requirements herein shall be as specified and fully described (see 6.2).

3.8 Standard equipment. All standard equipment normally provided with the manufacturer's commercial machine shall be furnished.

3.9 Optional equipment and accessories. Optional equipment and accessories shall be furnished as specified and fully described in the following sub-paragraphs.

3.9.1 Arbors, arbor supports and spacing collars. Arbors, arbor supports and spacing collars shall be provided in the styles and sizes specified (see 6.2).

3.9.2 Universal dividing head. A universal dividing head shall be furnished in the size specified (see 6.2). The dividing head shall be designed to index work or to rotate work in conjunction with the longitudinal movement of the table. The spindle shall be mounted on ball or roller bearings. The head stock shall be graduated in degrees to indicate the angle of inclination. When specified (see 6.2), gearing shall be provided to drive the dividing head from the table feed motor. Unless otherwise specified (see 6.2), the drive ratio between the crank and spindle shall be not more than 40 to 1. Unless otherwise specified (see 6.2), the indexing head shall be provided with the number of plates necessary for dividing a single rotation of the dividing head into each integral number of equal divisions between 2 and 50 inclusive, when using the plain indexing method. Additional plates shall be furnished as specified (see 6.2).

3.9.3 Rotary table. The rotary table shall have its base graduated in increments of not more than one degree. The diameter of the table shall be as specified (see 6.2).

3.9.4 Additional optional and accessory equipment. Additional optional and accessory equipment shall be furnished as specified and fully described

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(see 6.2). The equipment provided shall be functional without requiring modification of the equipment or machine.

3.10 Marking on instruments, control panels, charts, and plates. All words on instruments, control panels, charts, and plates shall be in the English language. Characters shall be permanently marked in boldface on a contrasting background. All plates shall be corrosion resistant.

3.10.1 Lubrication plate. When specified (see 6.2), a lubrication plate shall be attached to each machine. Unless otherwise specified (see 6.2), the information provided on the plate shall be as listed below:

Points of lubricant application
Servicing interval
Type of lubricant
Federal or military specification for each lubricant

3.10.2 Nameplate. A nameplate shall be securely attached to each machine. Unless otherwise specified (see 6.2), the nameplate shall contain the information listed below.

Nomenclature
Manufacturer's name
Manufacturer's model designation
Manufacturer's serial number
Power input (volts, total amps, phase, frequency)
Amp rating of largest motor
Short circuit/over-current protection rating
Contract number or order number
National Stock Number or Plant Equipment Code
Date of manufacture

3.10.3 Hydraulic fluid requirements plate. When a hydraulic unit is provided with the machine, a hydraulic fluid requirements plate shall be attached to the machine at the unit reservoir fill point. The plate shall contain the following information:

Type of fluid (SAE number or military or Federal specification number)
Type of filter

3.11 Workmanship. Workmanship of the milling machine and its accessories shall meet all requirements specified herein.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the

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contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any inspections set forth in the specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspections set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products and supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements; however, this does not authorize submissions of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.4).

b. Quality conformance inspection (see 4.5).

c. Acceptance test (see 4.6).

4.3 Inspection conditions. Unless otherwise specified (see 6 .2), all inspections shall be performed in accordance with the test conditions specified in 4.8.1.

4.4 First article inspection. When a first article inspection is required, it shall be applied to the first article submitted in accordance with 3.1. Unless otherwise specified (see 6.2), first article inspection shall consist of the examination in 4.7 and all tests in 4.8. The machine shall pass the examination and all tests to be accepted.

4.5 Quality conformance inspection. Quality conformance inspection shall be applied to each milling machine prior to being offered for acceptance under the contract. Unless otherwise specified (see 6.2), quality conformance inspection shall consist of the examination in 4.7, the tests in 4.8, and the inspection in 4.9. The machine shall pass the examination, all tests, and the inspection to be accepted.

4.6 Acceptance test. An acceptance test shall be performed on each machine to ensure conformance with this specification. Unless otherwise specified (see 6.2), the acceptance test shall be performed only after the milling machine is installed at its final location. The acceptance test shall consist of the examination in 4.7 and all tests in 4.8. The machine shall pass the examination and all tests to be accepted.

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4.7 Examination. Each milling machine shall be examined to determine compliance with all requirements of this specification.

4.8 Tests.

4.8.1 Test conditions. All tests shall be performed in an indoor facility with ambient conditions of 41° to 104° Fahrenheit and 15 to 95 percent relative humidity.

4.8.2 Operational test. Each milling machine shall be continuously operated at no load for not less than one hour. Proper operation of all controls, motors, adjusting mechanisms, and accessories shall be verified during the +trial period. If the milling machine does not pass the operational test in its entirety without failure, the test shall be repeated.

4.8.3 Alignment tolerance test. Each machine shall be tested to determine compliance with the alignment tolerance requirements of 3.6. The alignment tolerance test shall be performed prior to and subsequent to the performance test(s) given in 4.8.4 through 4.8.4.1. Any machine adjustments required to comply with 3.6 during or after the performance test shall be cause for rejection.

4.8.4 Performance test. A horsepower, milling accuracy and tool chatter test shall be performed on the vertical spindle and the horizontal. The vertical and the horizontal spindle on each machine shall be tested by machining a workpiece of SAE 4140 steel. The workpiece shall not be less than 16 inches long, 6 inches wide and 3 inches thick. The workpiece may be drilled and counter bored to facilitate mounting. One of the two surfaces having the largest area shall be ground flat and mounted in contact with the table. Roughing cuts shall be made to mill metal off the horizontal and one vertical surface of the workpiece. The roughing cuts shall be made at a sufficient depth and feed rate to load the spindle drive motor to not less than the maximum continous duty rated electrical input. The type of cutting tool, depth of cut, spindle speed and feed rate shall be determined by the manufacturer, provided the required loading is achieved. There shall be no evidence of chatter, stalling or permanent distortion during or after test. Finishing cuts shall be made on the horizontal and vertical surfaces. The finished surfaces shall meet the following accuracies:

4.8.4.1 Squareness. The two milled sides shall be tested for squareness. The sides shall be square with each other within 0.0010 inch per foot.

4.8.4.2 Flatness. The two milled sides shall be tested for flatness. The sides shall be flat within 0.0010 inch per foot.

4.8.4.3 Parallelism. The finished horizontal surface shall be tested for parallelism. The top surface shall be parallel to the bottom surface within 0.0010 inch per foot.

4.8.4.4 Climb milling test. Each milling machine shall be tested in accordance with the manufacturers standard climb milling test, provided that

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the test proves the capability of the machine to meet all of the requirements for which it was designed.

4.8.4.5 Additional performance tests. Performance tests required in addition to or in lieu of the requirements given in 4.8.4 shall be as specified and fully described (see 6.2).

4.8.5 Environmental protection certification of conformance. The contractor shall certify and maintain substantiating evidence that the product offered complies with all environmental protection restrictions as required in 3.2.6. The Government reserves the right to require proof of such conformance prior to first delivery and thereafter as may be otherwise provided for under the provisions of the contract.

4.8.6 Electromagnetic interference control test. Equipment requiring electromagnetic interference control shall be tested for compliance with 3.3.8 using the procedures given in MIL-STD-462.

4.9 Preservation/packaging inspection. Preservation and packaging of each item shall be inspected to determine compliance with the requirements of section 5.

4.10 Rough handling test. When rough handling test(s) are specified, they shall meet the the requirements of 5.4 and the ordering data in 6.2.

5. PACKAGING

5.1 Preservation/packaging. Preservation/packaging requirements shall be military level A or C or Commercial, as specified (see 6.2).

5.1.1 Military levels. Level A preservation/packaging shall be in accordance with the requirements as specified in MIL-M-18058. Level C shall be in accordance with the requirements as specified in MIL-STD-1190.

5.1.2 Commercial. Commercial shall be in accordance with ASTM D3951 as specified (see 6.2).

5.2 Packing. Packing requirements shall be military levels A, B or C, or commercial, as specified (see 6.2).

5.2.1 Military levels A, B or C. Levels A or B packing shall be in accordance with the requirements as specified in MIL-M-18058. Level C shall be in accordance with the requirements as specified in MIL-STD-1190 (see 6.2).

5.2.2 Commercial. Commercial packing shall be in accordance with ASTM D3951. However, it is recommended that commercial packing be used only when the item is to be used at its first destination (see 6.2).

5.3 Marking. All markings, military or commercial, for all interior or exterior containers shall be in accordance with the requirements as specified in MIL-STD-129.

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5.4 Rough handling test. When required, selected rough handling tests in MIL-M-18058 shall be performed as specified (see 6.2). All tests specified in MIL-M-18058 need not be performed. The completed unit pack, shipping container(s) and all contents shall be capable of withstanding the shipping and handling conditions expected from fork lift handling, crane handling and truck transportation. The container, its contents and the packaging shall not be damaged, functionally or physically, or items displaced by the appropriate rough handling test(s). The test(s) to be performed shall be either mechanical handling test or rotational drop test.

6. NOTES

6.1 Intended use. The milling machines covered by this specification are to be used for horizontal and vertical manual milling operations.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. Type, class, and size required (see 1.2).

c. Issue of DODISS to be cited in the solicitation and, if required, the specific issue of individual documents referenced (see 2.1.1 and 2.2).

d. First article, if required (see 3.1).

e. System of units for measuring and indicating devices (inch-pound or metric) (see 3.2.1).

f. Digital readout system, if not required (see 3.2.1.2).

g. Special digital readout system features, if required (see 3.2.1.2).

h. Special position feedback device features, if required (see 3.2.1.2.1)

i. Special electronic display console features, if required (see 3.2.1.2.2).

j. Additional safety and health requirements, if required (see 3.2.5).

k. Specify configuration of additional point-of-operation guarding, if required (see 3.2.5).

l. Specify environmental protection requirements exceeding EPA/ OSHA requirements (see 3.2.6).

m. Painting, if different (see 3.3.5).

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n. If electromagnetic interference control is required, specify the equipment and subsystem class and the emission and susceptibility requirements (see 3.3.8).

o. Knee elevating screw guard, if required (see 3.4.2).

p. Universal table mount, if required (see 3.4.3).

q. Special worktable, if required (see 3.4.7).

r. Power feed to quill, if required (see 3.4.8).

s. Power rapid traverse, if not required (see 3.4.10).

t. Electrical requirements, if different (see 3.4.16).

u. Initial voltage, as specified (see 3.4.16).

v. Motors, if different (see 3.4.16.1).

w. Worklight, if required (see 3.4.16.2).

x. Hour meter, if required (see 3.4.16.3).

y. Size and capacity, if different (see 3.5).

z. Specify physical size limitation, if required (see 3.5.1).

aa. Additional performance features, if required (see 3.7.1).

bb. Required styles and sizes of arbor supports, arbors and spacing collars (see 3.9.1).

cc. Size of dividing head required, driving gear, if required, drive ratio, if different, index plates, if different and additional index plates, if required (see 3.9.2).

dd. Required diameter of rotary table (see 3.9.3).

ee. Additional optional equipment, if required (see 3.9.4).

ff. Lubrication plate, if required (see 3.10.1).

gg. Lubrication plate information, if different (see 3.10.1).

hh. Nameplate, if different (see 3.10.2).

ii. Hydraulic plate, if required (see 3.10.3).

jj. Hydraulic plate informational, if different (see 3.10.3).

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kk. Inspection conditions, if different (see 4.3).

ll. First article inspection, if different (see 4.4).

mm. Quality conformance inspection, if different (see 4.5).

nn. Acceptance test, if different (see 4.6).

oo. Additional performance tests, if required (see 4.8.4.1).

pp. Selection of applicable level of preservation/packaging and packing (see 5.1 and 5.2).

qq. Selection of rough handling test requirements to be performed, if required (see 5.4).

6.3 Consideration of data requirements. The following data requirements should be considered when this specification is applied on a contract. The applicable Data Item Descriptions (DIDs) should be reviewed in conjunction with the specific acquisition to ensure that only essential data are requested/provided and that the DIDs are tailored to reflect the requirements of the specific acquisition. To ensure correct contractual application of the data requirements, a Contract Data Requirements List (DD Form 1423) must be prepared to obtain the data, except where DOD FAR Supplement 27.475-1 exempts the requirement for a DD Form 1423.

Reference Paragraph	DID Number	DID Title	Suggested Tailoring
N/A	DI-TMSS-80527	Commercial Off-The-Shelf (COTS) Manuals	Require all data be in the English
N/A	DI-TMSS-80528	Supplemental Data - Commercial Off-The-Shelf (COTS) Manuals	Require all data be in the English language

The above DIDs were those cleared as of the date of this specification. The current issue of DOD 5010.12-L, Acquisition Management Systems and Data Requirements Control List (AMSDL), must be researched to ensure that only current, cleared DIDs are cited on the DD Form 1423.

6.3.1 Technical data. It should be specified in the "Remarks" section of the DD Form 1423 that all technical data ordered shall be written in the English language.

6.4 First article. When first article inspection is required, the contracting officer should provide specific guidance to offerors whether the item(s) should be a first article sample, a first production item, or a standard production item from the contractor's current inventory and the number of items to be tested as specified in 4.4. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, approval of first article test results, and disposition of first articles. Invitations for bids should provide that the Government

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reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract.

6.5 Definitions.

6.5.1 Inch-pound units. Inch-pound units are a system of measures based on the yard and pound commonly used in the United States of America and defined by the National Institute of Standards and Technology. Inch-pound units having the same names in other countries may differ in magnitude.

6.5.2 Metric units. Metric units are a system of basic measures defined by the International System of Units based on "Le Systeme International D'Unites (SI)," of the International Bureau of Weights and Measures. These units are described in ASTM E 380 and IEEE 268.

6.6 Measurement system. In this specification, all measurements, dimensions, sizes, and capacities are given in inch-pound units. These measurements may be converted to metric units through the use of the conversion factors and methods specified in FED-STD-376.

6.7 Cross-reference of classification changes. Classification changes from the previous issue of this specification are as follows:

a. Type I, Style A, added Size 4.
b. Type I, Style B, added Size 2.

6.8 Changes from previous issues. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

6.9 Safety and health requirements. OSHA 29 CFR 1910 limits only the total hazard level (noise, radiation, electromagnetic emissions, noxious vapors, air contaminants, and heat) of the environment in which a machine will operate. It does not limit the hazard level of individual machines in an operating environment. The procuring activity should analyze the existing hazard level in the proposed operating environment and specify additional requirements necessary to integrate this new machine into its future environment.

6.10 Hazardous materials minimization. The manufacturer should minimize the use, generation, or emission of materials known hazardous to the environment during the manufacturing processes employed at his facilities. During the production of an end item, the manufacturer should comply with all applicable Federal, state, and local statutes in effect at the point of production which regulate the use of hazardous materials. Where practical, alternative materials or processes should be employed in lieu of known hazardous materials or processes to minimize the threat to the environment. Design criteria shall be evaluated to minimize the hazardous waste generated in the manufacture, use, and disposal of the milling machine.

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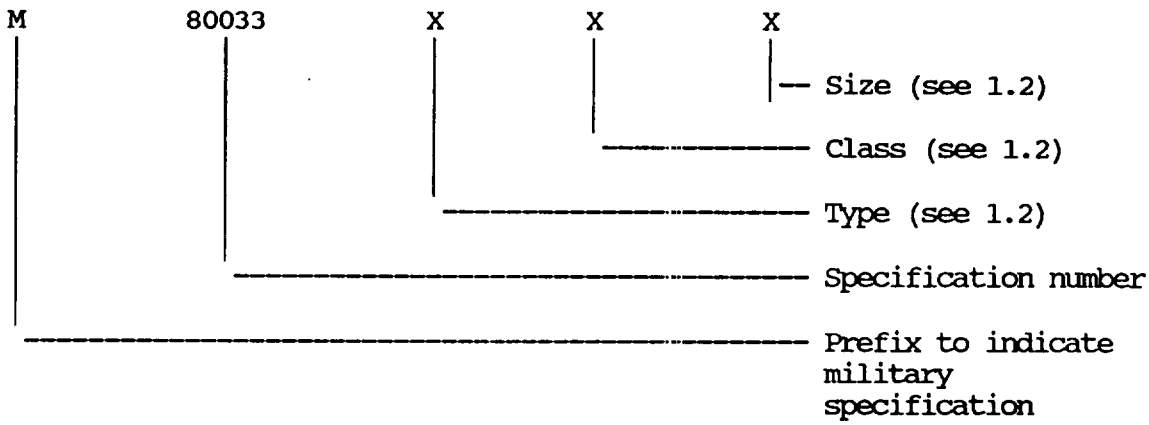
6.11 Training. Training required by the procuring activity shall be provided as specified in the contract.

6.12 Warranty. Warranty requirements shall be as specified by the procuring activity in the contract.

6.13 Inspection location. The contractor shall identify in his response to the solicitation the location where inspection and tests are to be performed.

6.14 Spare parts. Spare parts requirements (parts subject to normal wear and replacement such as belts or light bulbs) should be determined by the procuring activity and specified in the contract. Quantities of many spare parts may already be available at an installation and will not need to be purchased.

6.15 Part or Identifying Number (PIN). The PIN to be used for machines acquired to this specification are created as follows:



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TABLE I. TYPE I, DUAL SPINDLE TYPE MACHINES

CHARACTERISTICS	STYLE A TURRET POSITIONING of DESIRED SPINDLE				STYLE B RAM MOUNT VERTICAL INT. HOR. SPINDLE	
	SIZE 1	SIZE 2	SIZE 3	SIZE 4	SIZE 1	SIZE 2
Spindle HP, vertical	1	2	3	8	2	5
Spindle HP, horizontal	1	2	5	8	2	5
Table travel -						
Longitudinal (X)	21	22	28	38	30	34
Cross (Y)	13	10	11	16	12	12
Vertical (Z)	15	17	16	27	16	18
Table size, (L x W)	36 x 8	42 x 10	48 x 10	49 x 11	42 x 8	49 x 11
Spindle nose taper, size	30 ANS R-8	40 ANS 30 ANS	50 ANS 30 ANS	40 NMT 30 NMT	40 ANS R-8	40 NMT 30 NMT
Spindle speeds -						
Horizontal, no. of	6	8	infinite	18	infinite	
Speed range, RPM	185 to 2500	100 to 1130	29 to 1450	47 to 1900	50 to 3500	35 to 1800
Vertical, no. of	6	infinite	6	18	infinite	
Speed range, RPM	185 to 2500	100 to 3600	185 to 2500	47 to 1800	50 to 3500	35 to 1800
Table feeds, no. of	infinite	9	12	12	infinite	infinite
Range of table feeds, IPM						
Horizontal	1 to 18	3/4 to 15	3/4 to 28	1/4 to 21	3/4 to 25	3/8 to 24
Cross	----------	3/4 to 15	3/4 to 28	1/4 to 21	3/4 to 25	3/8 to 24
Vertical	----------	----------	3/8 to 14	1/4 to 21	3/4 to 25	3/8 to 24
Quill movement, vertical spindle	4	5	5-1/2	4-3/4	4	5-1/2
Power rapid traverse, IPM						
Longitudinal	----------	----------	80	----------	40	78
Cross	----------	----------	80	----------	40	78
Vertical	----------	----------	40	----------	15	27

NOTE: Sizes and capacities shall not be less than specified, but may exceed the specified sizes and capacities by not more than 10 percent provided that the 10 percent increase does not allow the next size machine to be furnished for the original requirement.

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TABLE I. (continued) - TYPE II, SINGLE SPINDLE TYPE MACHINES

CHARACTERISTICS	STYLE A, SINGLE SWIVEL DUAL RAM, SWIVEL PARALLEL TO RAM MOVEMENT SIZE 4	STYLE B, DUAL SWIVEL SPINDLE ON RAM SIZE 1
Spindle horsepower	10	5
Table travel -		
Longitudinal (X)	34	36
Cross (Y)	14	8
Vertical (Z)	23	16
Table size (L x W)	64 X 14	46 X 16
Spindle nose taper, ANS horizontal taper	50	40
Spindle speeds, no. of,		
Horizontal spindle	12	12
speed range, RPM	50 to 2000	50 to 1600
Table feeds, no. of	18	9
Feed range, IPM -		
Longitudinal (X)	3/8 to 36	3/4 to 20
Cross (Y)	5/16 to 24	3/4 to 20
Vertical (Z)	5/32 to 12	----------
Power rapid traverse, IPM -		
Longitudinal (X)	100	----------
Cross (Y)	75	----------
Vertical (Z)	40	----------
Quill movement	----------	----------
Feed motor, horse power	2	----------

NOTE: Sizes and capacities shall not be less than specified, but may exceed specified sizes and capacities by not more than 10 percent provided that the 10% increase does not allow the next size machine to be furnished for the original requirement.

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TABLE II. ALIGNMENT TOLERANCES

DESCRIPTION	ALLOWABLE DEVIATION NOT MORE THAN
Bore of arm bracket aligned with spindle	0.002 inch TIR
Spindle end periphery run out	0.0005 inch TIR
Spindle end face run out	0.0005 inch TIR
Spindle internal taper run out	
1 inch from spindle nose	0.0004 inch TIR
12 inches from spindle nose	0.0010 inch TIR
Table top parallel with horizontal spindle, low point at rear of table	0.0010 inches in 12 inches
Table top square with vertical spindle	0.0010 inches in 12 inches
Transverse movement of table parallel with axis of horizontal spindle	0.0010 inches in 12 inches
Worktable square with column knee ways tested with the knee clamped	
In plane through column, low point at rear of table	0.0010 inches in 12 inches
In plane parallel to ways	0.0010 inches in 12 inches
Rise and fall of table in longitudinal movement	0.0005 inches in 12 inches
T-slots parallel with horizontal spindle	0.0010 inches in 12 inches
T-slots parallel with table movement	0.0010 inches in 12 inches
Turret movement parallel with table surface on radius of vertical spindle	0.0010 inches in 12 inches
Ram movement parallel with table surface as determined at spindle nose	0.0010 inches in 12 inches

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CUSTODIANS:

Army - AL
Navy - SH
Air Force - 99

PREPARING ACTIVITY:

DLA - IP

PROJECT NUMBER:

3417 - 0191

REVIEW ACTIVITIES:

Navy - SH, YD
Air Force - 84
DLA - GS

USER ACTIVITY:

Navy - MC

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER	2. DOCUMENT DATE (YYMMDD)
	MIL-M-80033D	92 APRIL 21

3. DOCUMENT TITLE
MILLING MACHINES, COMBINATION HORIZONTAL AND VERTICAL, KNEE TYPE

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME	b. TELEPHONE (Include Area Code)	
DEFENSE INDUSTRIAL PLANT EQUIPMENT CENTER ATTN: DIPEC-SSM	(1) Commercial (901) 775-4985	(2) AUTOVON DSN 683-4985
c. ADDRESS (Include Zip Code) 2163 AIRWAYS BLVD. MEMPHIS, TN 38114-5213	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340	

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